

Data-Driven Cognitive Phenotyping in Acquired Brain Injury

Two stable cognitive phenotypes that survive the choice of imputation method

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Part 1

Background

Same injury severity, very different cognitive profiles — severity labels hide what should guide treatment

- **Acquired brain injury** (stroke, TBI, hypoxia, encephalitis) is a leading cause of long-term disability in working-age adults.

Effects scatter across attention, memory, language, executive function and visuoperception.

Diagnosis and severity bands group together patients whose cognitive profiles differ substantially.

The within-group variation that could guide rehabilitation stays hidden.

The question

Can data-driven clustering recover clinically meaningful cognitive phenotypes that diagnosis alone misses — and can we trust them?

Four pre-registered questions, each with a quantitative threshold

- | | | |
|----------|---|---|
| 1 | Distinct phenotypes? Does clustering find discrete cognitive phenotypes? | H1 Silhouette > 0.40, noise < 30%, ≥ 2 clusters |
| 2 | Imputation-robust? Are phenotypes stable across imputation methods? | H2 Mean pairwise ARI > 0.50 |
| 3 | Clinically linked? Does phenotype relate to the treating clinical unit? | H3 χ^2 significant, Cramér's V > 0.15 |
| 4 | Better at domain level? Do 6 cognitive domains beat 15 raw tests? | H4 Domain wins ≥ 3 of 4 quality metrics |

Part 2

Data & Methods

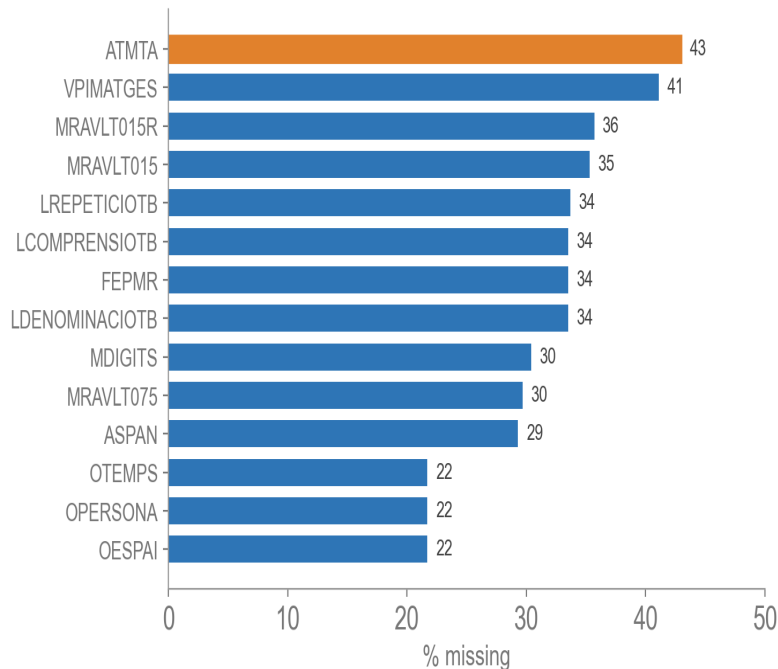
Routine clinical data is large but 58% incomplete — imputation is unavoidable, not optional

- **22,075 assessments** from 8,739 patients, one rehabilitation centre.

15 cognitive tests → **6 domains** after a 50%-missing filter.

Only 41.9% of rows are complete — listwise deletion would discard 58% of the data.

Pipeline



Ten imputation methods across five categories — extending the standard taxonomy with deep learning

Category	Methods	Idea
Conventional statistical	Mean · EM · MICE	distributional / chained equations
Machine learning	KNN · MissForest	non-parametric, non-linear
Hybrid	PMM	draws real donor values
Matrix completion	SoftImpute · NMF	low-rank reconstruction
Deep learning (new)	DAE · VAE	learn non-linear latent structure

10 methods × 2 representations (variable- and domain-level) = 20 pipeline runs. DAE/VAE detail in Appendix B.

A 1,080-configuration sweep selects clusters by a single quality score — no hand-tuning



UMAP grid

- n_neighbors: 15, 30, 50
- n_components: 3, 5, 8
- min_dist: 0.0

HDBSCAN grid

- min_cluster_size: 500–3000
- min_samples: 5, 10, 25, 50
- method: eom / leaf · ϵ : 0.0, 0.1, 0.5

Selection score $Q = \text{Silhouette} \times (1 - \text{noise proportion})$

For H2, a shared pipeline isolates imputation as the only varied factor

1. Fit StandardScaler on MICE → transform every other method.
2. Fit UMAP on MICE → transform every other method.
3. Fit K-Means on the MICE embedding → predict all others.

Same k, same boundaries, same embedding geometry. The only thing that changes between runs is the imputed values themselves — so any difference in cluster assignment is attributable to imputation alone.

HDBSCAN labels are kept for H1/H3/H4; K-Means labels are used only for the H2 concordance test.

Part 3

Results

H1 ✓ Two stable cognitive phenotypes emerge

15,072

Near-Normal

comparatively preserved within ABI

7,003

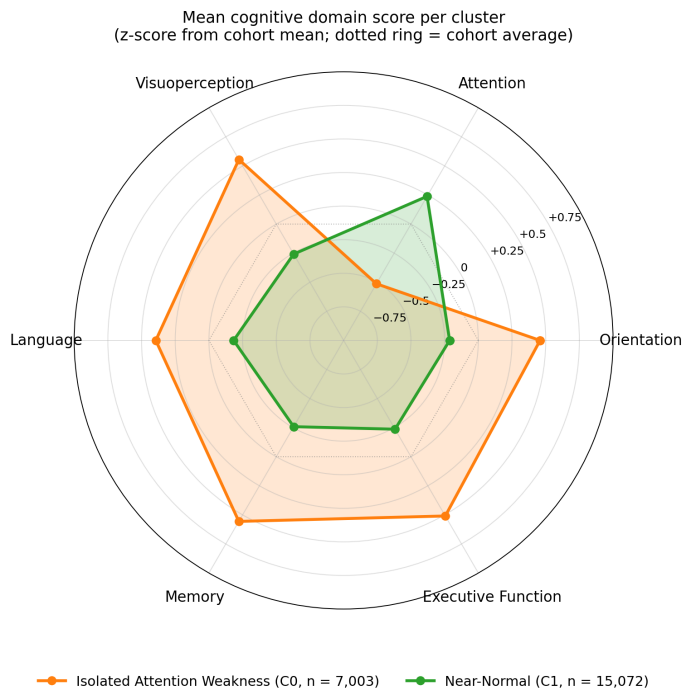
Isolated Attention Weakness

selective attention deficit, rest intact

- **Found by 8 of 10 imputation methods** (silhouette > 0.40 , noise $< 30\%$); MICE solution is the reference (silhouette 0.49, $k = 2$).

Per-cluster core stability (Hennig Jaccard) ≈ 0.63 for both phenotypes — approaching the 0.70 convention.

The two phenotypes are clinically coherent: strong-but-for-attention vs. uniformly near-average



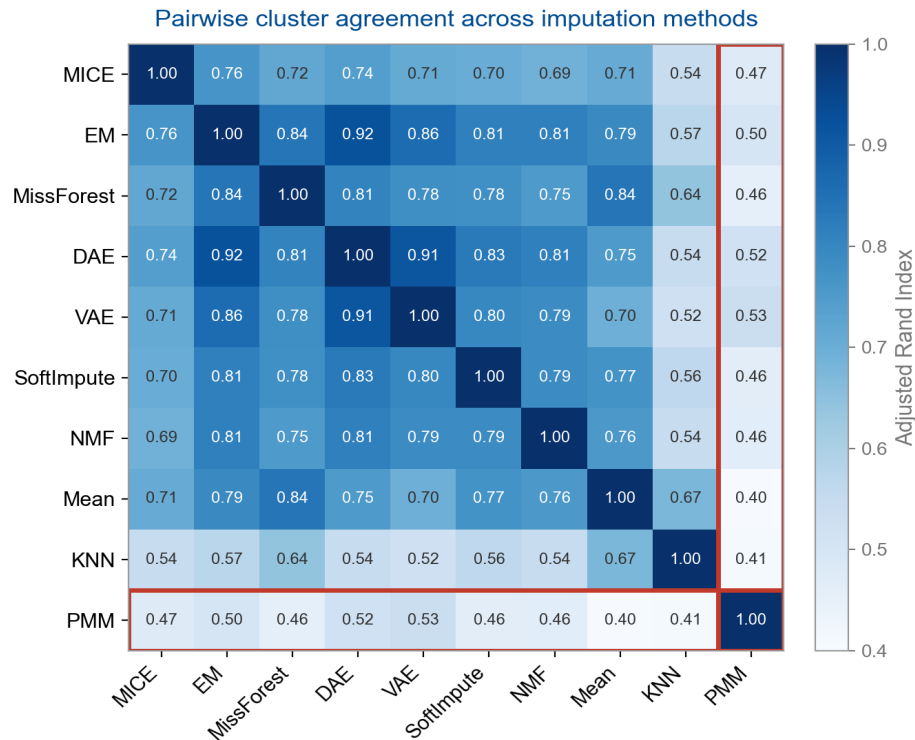
What to take away

- **Orange (IAW):** above the cohort mean everywhere except attention, which drops sharply — a selective deficit.

Navy (Near-Normal): flat, close to the cohort average across all six domains.

Between-cluster separation is large (Cohen's $d = 0.6$ – 0.9 per domain).

H2 ✓ Phenotypes are robust to imputation choice — only PMM breaks agreement



What to take away

- **Mean pairwise ARI = 0.68** [95% CI 0.68–0.69] — well above the 0.50 threshold.

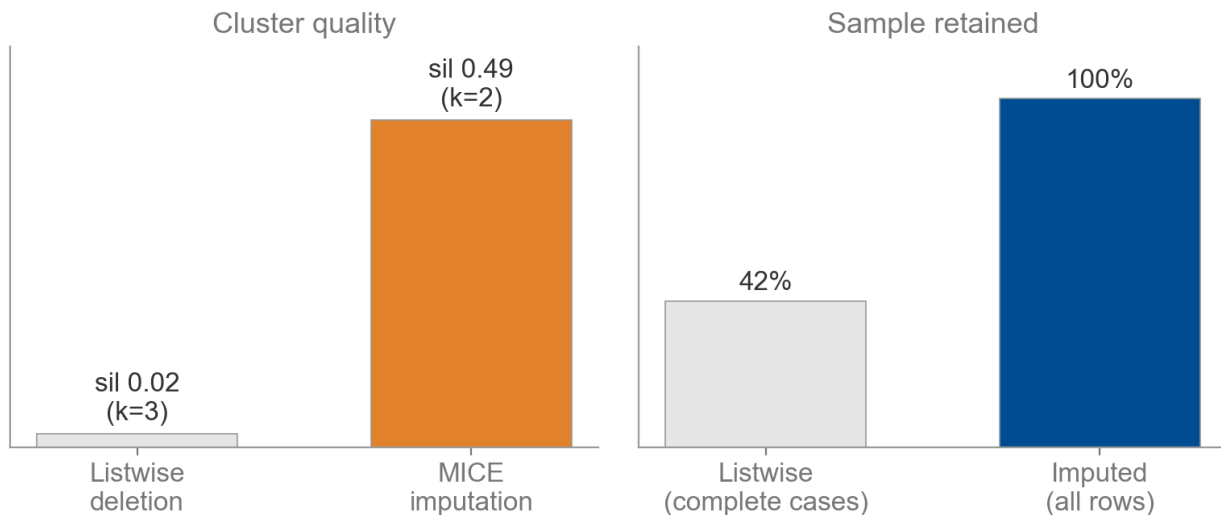
38 of 45 method pairs pass after Holm correction; all 7 failures involve PMM (red).

92% of patients receive a high-confidence label under consensus clustering.

Practical guidance: use any method except PMM.

Imputation is not cosmetic: discarding incomplete rows changes the cluster structure itself

Listwise deletion: more clusters, far weaker structure (ARI 0.62 vs MICE)



Why it matters

- Listwise deletion keeps only 42% of rows.
- It returns 3 clusters with silhouette 0.02 — essentially no structure.
- **ARI vs MICE = 0.62: a different partition, not a subset.**
- **The choice to impute is itself a finding.**

H3 ✓ Phenotype tracks the treating clinical unit — and the link survives adjustment for diagnosis

$$\chi^2 = 635$$

association of phenotype with clinical unit ($p < 0.001$)

$$V = 0.17$$

Cramér's V effect size — above the 0.15 bar

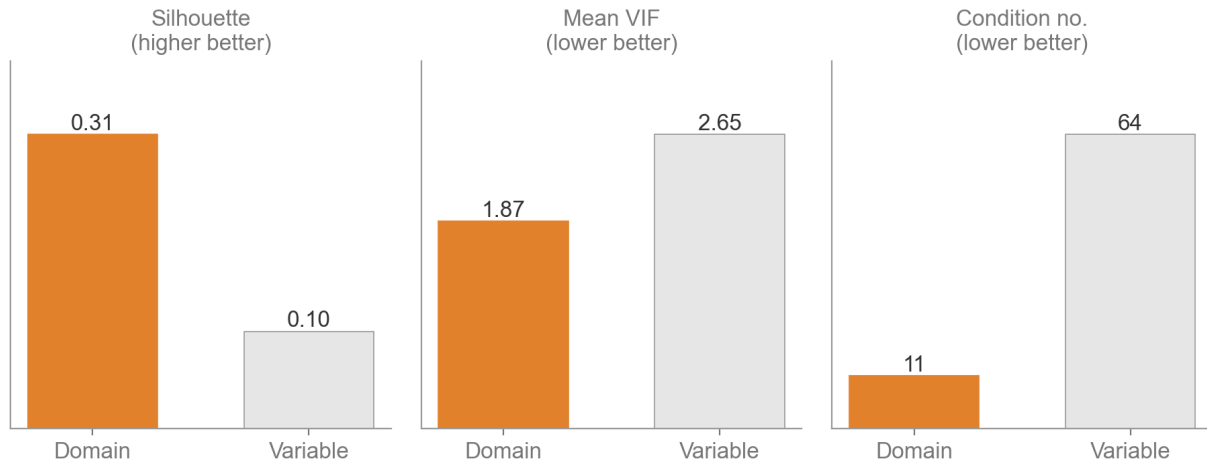
$$p = 2 \times 10^{-14}$$

Cochran–Mantel–Haenszel, stratified by diagnosis

- **The association is not a diagnosis confound:** it holds after stratifying by diagnosis group, so phenotype adds explanatory power beyond the diagnostic label — supporting phenotype-informed routing of patients.

H4 ✓ Six domains beat fifteen raw tests on 3 of 4 metrics

Domain-level features win 3 of 4 quality metrics



Why it matters

- **Cleaner separation** and far less multicollinearity.

More interpretable — clinicians reason in domains.

Variable-level wins only on noise % (the 4th metric, not shown).

All four pre-registered hypotheses are supported

Hypothesis	Verdict	Key evidence
H1 Discrete phenotypes	Supported	8/10 methods pass silhouette + noise; core Jaccard ≈ 0.63
H2 Imputation robustness	Supported	Mean ARI 0.68; 38/45 pairs survive Holm correction
H3 Clinical-unit link	Supported	$\chi^2 = 635$, $V = 0.17$, CMH $p = 2 \times 10^{-14}$ (diagnosis-adjusted)
H4 Domain vs variable	Supported	Domain wins 3 of 4 quality metrics

Robustness: all four conclusions hold under bootstrap 95% CIs, Holm correction, Rubin pooling ($M = 5$), and a listwise-deletion baseline.

Part 4

Discussion & Conclusions

Both phenotypes match cognitive patterns already described in the ABI literature

- **Attention is a recognised vulnerability after ABI**
— processing-speed and attention deficits are among the most consistent findings in stroke and TBI cohorts.

An isolated-attention profile with otherwise preserved domains is clinically familiar; our data recover it without supervision.

A near-normal majority is expected in a routine rehabilitation sample spanning the recovery continuum.

Contribution: the structure is data-driven and reproducible, not imposed by clinical expectation.

The signature

Isolated Attention Weakness

= high on every domain except a sharp attention dip.

Between-cluster Cohen's $d = 0.6\text{--}0.9$.

Phenotypes translate into concrete, differentiated rehabilitation pathways

For the clinic

- Near-Normal → vocational rehabilitation and community-return planning.
- Attention Weakness → attention retraining and structured, low-distraction settings.
- Report each assignment with its confidence, not as a hard label.

For the analyst

- Use MICE as the default for clinical neuropsychology data.
- Test sensitivity with at least two imputation methods.
- Cluster on domain-level features, not raw tests.

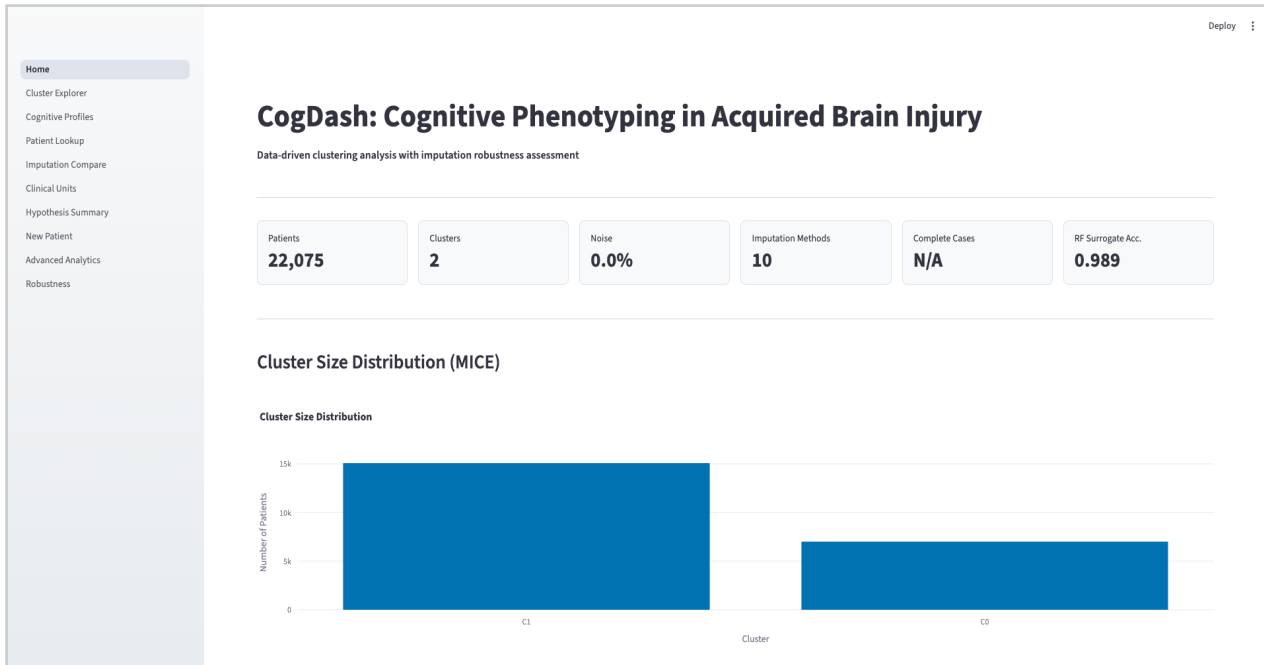
One framework, two audiences: phenotypes guide bedside decisions and set the defaults for the next analysis.

Phenotypes power CogDash — decision support that classifies a new patient and reports its confidence

8 modules:

cluster explorer,
cognitive profiles,
patient lookup,
imputation compare,
clinical units, hypothesis
summary, new-patient
classifier, advanced
analytics.

*Streamlit + Plotly,
colour-blind-safe,
Docker-packaged.*



Main limitations are design constraints — partly offset by built-in sensitivity analyses

Single centre — but the test battery (TMT, Stroop, digit span, RAVLT) is internationally standard

Cross-sectional — yet multiple assessments per patient span the recovery continuum

MAR assumption — 10 methods, Rubin pooling and a listwise baseline serve as built-in sensitivity checks

No functional outcomes — (return-to-work, length of stay); only outcome-adjacent proxies available

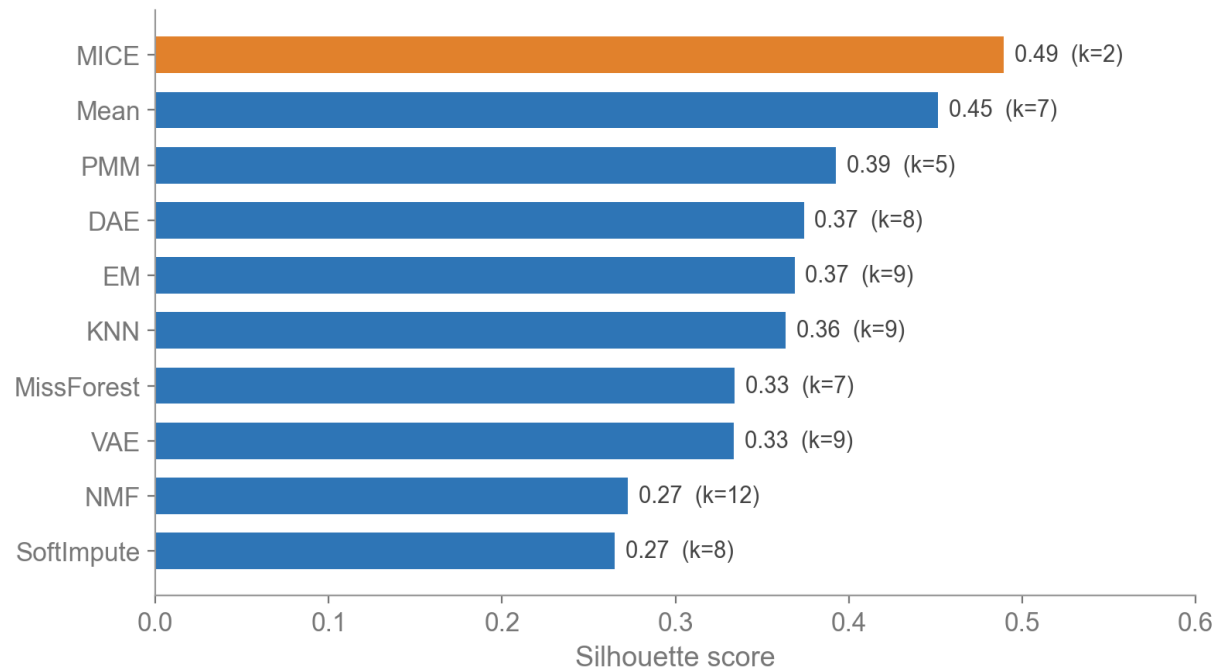
Conclusions

- 1. Two robust cognitive phenotypes in ABI** — Near-Normal and Isolated Attention Weakness — cutting across diagnostic categories.
- 2. Imputation is not cosmetic** — listwise deletion changes the structure; use any method except PMM.
- 3. Phenotype adds clinical value beyond diagnosis** (CMH $p = 2 \times 10^{-14}$); 92% of patients get high-confidence labels.
- 4. A transferable imputation-robustness framework** for clustering any dataset with substantial missingness.

References

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Cluster quality varies with imputation, but the two-phenotype signal persists



- Silhouette and k differ across methods (k = 2–12).
- **MICE gives the cleanest 2-cluster solution.**
- Matrix-completion methods (SoftImpute, NMF) are weakest.

Two targeted fixes make autoencoder imputation work on structured missingness

DAE — pattern-aware corruption

- Real missingness is structured: tests are ordered together.
- 50% of training masks sampled from the empirical co-occurrence pattern.
- Architecture $15 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow \dots \rightarrow 15$; 5 runs averaged.

VAE — cyclical β -annealing

- Fixed β causes posterior collapse — latent space ignored.
- β ramps $0 \rightarrow 0.1$ over 40-epoch cycles.
- Latent dim 32; $K = 20$ samples averaged.

Held-out 10% RMSE confirms DAE outperforms VAE on every variable.

Every headline result was re-tested under stricter conditions

Re-estimation

- Bootstrap 95% CIs ($B = 1000$): ARI [0.68, 0.69]; V [0.16, 0.19].
- Holm correction over 45 H2 pairs: 38 survive; 7 failures all involve PMM.
- Rubin pooling ($M = 5$ MICE): conclusions unchanged.

Baselines & stability

- Listwise baseline: 3 clusters, silhouette 0.02 vs 0.49 — imputation matters.
- 68.9% of multi-assessment patients keep their phenotype over time.
- Time-since-injury differs by phenotype (214 vs 163 days, $p \approx 10^{-80}$).